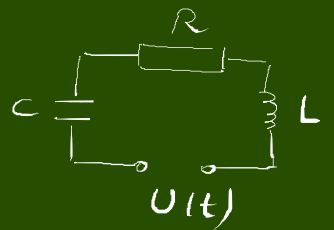
Ecuații diferențiale liniare de ordin superior cu coeficienți constanți

Exemplu. RLC serie



$$U(t) = U_C(t) + U_R(t) + U_L(t)$$

$$\begin{cases} U_R(t) = I(t)\,R \quad (\text{Ohm}) \\ U_L(t) = L \cdot I'(t) \quad (\text{L autoinductie}) \\ I(t) = Q'(t) \\ Q(t) = \frac{1}{C} U_C(t) \end{cases}$$

$$\Rightarrow \quad \underbrace{\frac{1}{C} Q(t) + R \cdot Q'(t) + L \cdot Q''(t)}_{\text{ec. dif. lin. de ordinul 2}} = U(t)$$

Definiție Ecuație diferențială liniară de ordin n

$$x^{(n)} + a_1 x^{(n-1)} + \dots + a_{n-1} x' + a_n x = b(t) \quad (1)$$

$x' \ \frac{dx}{dt}$

t variabilă

$x = x(t)$ fct. nec. (derivabilă de n ori)

$x'' \ \frac{d^2x}{dt^2}$

$x', \dots, x^{(n-1)}, x^{(n)}$ derivatele fct. nec. $x(t)$; $x^{(n)} \quad \frac{d^n x}{dt^n}$

$a_1 \dots a_n \in \mathbb{R}$

$b : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}$ fct. continuă

$b(t) = 0 \ \forall t \Rightarrow$ ec. dif. omogenă

$b(t) \neq 0 \Rightarrow$ ec. dif. neomogenă

(2)

Condiții inițiale pt. ec dif. (1) : $x(t_0) = x_0, x'(t_0) = x_1 \dots x^{(n-1)}(t_0) = x_{n-1}$

$t_0 \in I, x_0 \dots x_{n-1} \in \mathbb{R}$.

Obs. Fie $x(t)$ o soluție a ec. dif. (1)

Notăm cu $y_1 = x$

$y_2 = x'$

$\vdots$

$y_n = x^{(n-1)}$

$\Rightarrow$

$$\begin{cases} y_1' = y_2 \\ y_2' = y_3 \\ \vdots \\ y_{n-1}' = y_n \\ y_n' = -a_n y_1 - a_{n-1} y_2 - \dots - a_1 y_n + b(t) \end{cases} \quad (3)$$

$\Rightarrow$ $x(t)$ sol. a ec. dif. (1) $\Rightarrow \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$ sol a sist. de ec. dif (3)

$\Leftarrow$

fie $\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$ sol. a sist. de ec. dif (3)

Notăm $x = y_1$

$\Rightarrow x' = y_1' = y_2$

$x'' = y_2' = y_3$

$\cdots\cdots$

$x^{(n-1)} = y_n$.

$x^{(n)} = -a_n \cdot y_1 - \cdots - a_1 y_n + b(t) = -a_n \cdot x - \cdots - a_1 x^{(n-1)} + b(t)$

$\Rightarrow x^{(n)} + a_1 x^{(n-1)} + \cdots + a_n \cdot x = b(t) \Rightarrow x$ sol. a ec. dif (1)

Propoziție 1) Problema Cauchy $\begin{cases} x^{(n)} + a_1 x^{(n-1)} + \cdots + a_n x = b(t) & (1) \\ x(t_0) = x_0 \ \cdots \ x^{(n-1)}(t_0) = x_{n-1} & (2) \end{cases}$

are soluție unică $\forall\ t_0 \in I,\ x_0 \cdots x_{n-1} \in \mathbb{R}$

(1) $\Longleftrightarrow$ (3)

(2) $\Longleftarrow$?

$$\begin{cases} y_1 = x \\ y_2 = x' \\ \vdots \\ y_n = x^{(n-1)} \end{cases} \Longleftrightarrow \begin{cases} y_1(t_0) = x(t_0) = x_0 \\ y_2(t_0) = x'(t_0) = x_1 \\ \text{---} \\ y_n(t_0) = x^{(n-1)}(t_0) = x_n \end{cases}$$

C.I. pt Hoskew (4)

Caso particolare $b(t) = 0$ (ec. diff. omogenea)

Pb. Cauchy
$$\begin{cases} x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x = 0 \quad (5) \\ x(t_0) = 0 \dots x^{(n-1)}(t_0) = 0 \end{cases}$$
ave soluzione unica $x(t) = 0 \ \forall t \in \mathbb{R}$

2) $\mathcal{S}_0 = \{ x : \mathbb{R} \longrightarrow \mathbb{R} \mid x$ derivabile n volte e verifica (5) $\}$

sottosp. sol. ec diff. lin omogenee

$S_0 = \{ Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} : \mathbb{R} \longrightarrow \mathbb{R}^n \mid Y$ deriv. e $Y' = AY \}$

dove $A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ 0 & 0 & 0 & \dots & 1 \\ -a_n & -a_{n-1} & \dots & & -a_1 \end{pmatrix}$ sp. vettoriale di dim. n.

$\Rightarrow$ $\mathcal{S}_0$ spațiu vectorial izomorf cu S_0

($\dim_{\mathbb{R}} \mathcal{S}_0 = n$)

3) $\mathcal{S}_{n0} = \{ x : I \longrightarrow \mathbb{R} \mid x$ derivabilă de n ori și x verifică (1) $\}$

fie $x_p \in \mathcal{S}_{n0}$ o sol. particulară.

Atunci $x \in \mathcal{S}_{n0} \iff x - x_p \in \mathcal{S}_0$

($\mathcal{S}_{n0} = \mathcal{S}_0 + \{x_p\}$)

$x = x_0 + x_p$

<u>Consecință</u> Dacă $x_1(t), \dots, x_n(t)$ bază în $\mathcal{S}_0$ (sistem fundamental de soluții) $\Rightarrow$ sol. generală a ecuației dif. omogene (5) este

$x(t) = C_1 x_1(t) + \dots + C_n x_n(t)$, $C_1 \dots C_n \in \mathbb{R}$.

$n = 2$ ecuația diferențială liniară omogenă de ordinul 2

$$x'' + ax' + bx = 0 \quad (6) \qquad a, b \in \mathbb{R}$$

Caut o soluție a ec. dif. (6) de forma $x(t) = e^{\lambda t}$

$x'(t) = \lambda e^{\lambda t}$

$x''(t) = \lambda^2 e^{\lambda t}$

$$\Rightarrow \lambda^2 e^{\lambda t} + a\lambda e^{\lambda t} + b e^{\lambda t} = 0 \Rightarrow$$

$$\Rightarrow \boxed{\lambda^2 + a\lambda + b = 0}$$ ecuația caracteristică asociată ecuației diferențiale

$\lambda_1, \lambda_2 \in \mathbb{R}$, $\lambda_1 \neq \lambda_2$ | $\lambda_1 = \lambda_2 \in \mathbb{R}$ | $\lambda_1, \lambda_2 \in \mathbb{C} - \mathbb{R}$, $\lambda_2 = \overline{\lambda_1}$

$x'' + ax' + bx = 0 \longrightarrow$ $y_1 = x$, $y_2 = x'$

$$\begin{cases} y_1' = y_2 \\ y_2' = -by_1 - ay_2 \end{cases}$$

$$A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}$$

$$P_A(X) = X^2 + aX + b$$

I $\lambda_1 \neq \lambda_2$, $\lambda_1, \lambda_2 \in \mathbb{R}$

$\Rightarrow x_1(t) = e^{\lambda_1 t}$, $x_2(t) = e^{\lambda_2 t}$ soluții ale ec. diff. (6)

x_1, x_2 fct. liniar independente — direct, în definiția

$\mathcal{S}_0 \simeq S_0$

x_1, x_2 liniar indep. $\Leftrightarrow \begin{pmatrix} x_1 \\ x_1' \end{pmatrix}, \begin{pmatrix} x_2 \\ x_2' \end{pmatrix}$ liniar indep. $\Leftrightarrow \begin{pmatrix} x_1(t_0) \\ x_1'(t_0) \end{pmatrix} \begin{pmatrix} x_2(t_0) \\ x_2'(t_0) \end{pmatrix}$ vectori din $\mathbb{R}^2$ liniar indep. ✓

Fie $t_0 = 0$

$\Rightarrow \begin{pmatrix} x_1(t_0) \\ x_1'(t_0) \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix}$, $\begin{pmatrix} x_2(t_0) \\ x_2'(t_0) \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix}$

$\begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix}, \begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix}$ liniar indep. $\Leftrightarrow \begin{vmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{vmatrix} \neq 0 \Leftrightarrow \lambda_2 - \lambda_1 \neq 0$ ✓

$\Rightarrow$ $x_1(t)$, $x_2(t)$ baza în $\mathcal{S}_0$

$\Rightarrow$ sol. generală $x(t) = C_1 x_1(t) + C_2 x_2(t)$, $C_1, C_2 \in \mathbb{R}$

exemplu: $x'' - 2x' - 8x = 0$ $\quad x(0) = 5,\ x'(0) = 2$

$\lambda^2 - 2\lambda - 8 = 0$ $\quad \lambda_1 = 4 \Rightarrow x_1(t) = e^{\lambda_1 t} = e^{4t}$

$\lambda_2 = -2 \Rightarrow x_2(t) = e^{\lambda_2 t} = e^{-2t}$

$\Rightarrow$ sol generală a ec. dif. $x(t) = C_1 x_1(t) + C_2 x_2(t)$

$= C_1 e^{4t} + C_2 e^{-2t}$ $\quad C_1, C_2 \in \mathbb{R}$

$$\begin{cases} x(0) = 5 \\ x'(0) = 2 \end{cases} \Rightarrow \begin{matrix} C_1 + C_2 = 5 \quad | \cdot 2 \\ 4C_1 - 2C_2 = 2 \end{matrix} \Rightarrow \begin{matrix} C_1 = 2 \\ C_2 = 3 \end{matrix}$$

$\Rightarrow \quad x(t) = 2e^{4t} + 3e^{-2t}$

II $\quad \lambda_1 = \lambda_2 \in \mathbb{R}$ $\quad$ not $\lambda = \lambda_1 = \lambda_2$

$\Rightarrow x_1(t) = e^{\lambda t}$ sol. a ec. dif. $x'' + ax' + bx = 0$

fie $x_2(t) = te^{\lambda t}$ $\quad x_2$ sol. a ec. dif. ↑

x_1, x_2 sol. liniar indep.

$x_2(t) = te^{\lambda t}$

$x_2'(t) = e^{\lambda t} + t\lambda e^{\lambda t}$

$x_2''(t) = 2\lambda e^{\lambda t} + t\lambda^2 e^{\lambda t}$

$\Rightarrow e^{\lambda t}(2\lambda + t\lambda^2) + ae^{\lambda t}(1 + t\lambda) + bte^{\lambda t} \stackrel{?}{=} 0$

$e^{\lambda t}\big(t\underbrace{(\lambda^2 + a\lambda + b)}_{0} + \underbrace{(2\lambda + a)}_{0}\big) \stackrel{?}{=} 0$

$\Rightarrow x_2(t)$ sol. a ec. dif.

Linia independenta, x_1, x_2 lin. indep $\Leftrightarrow \begin{pmatrix} x_1 \\ x_1' \end{pmatrix}, \begin{pmatrix} x_2 \\ x_2' \end{pmatrix}$ lin. indep $\Leftrightarrow$

$\Leftrightarrow \begin{pmatrix} x_1(0) \\ x_1'(0) \end{pmatrix}, \begin{pmatrix} x_2(0) \\ x_2'(0) \end{pmatrix} \in \mathbb{R}^2$ lin. indep

" "

$\begin{pmatrix} 1 \\ \lambda \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\begin{pmatrix} 1 \\ \lambda \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ lin. indep $\Leftrightarrow \begin{vmatrix} 1 & 0 \\ \lambda & 1 \end{vmatrix} = 1 \neq 0$

$\Rightarrow$ $x_1(t), x_2(t)$ baza în $\mathcal{S}_0$

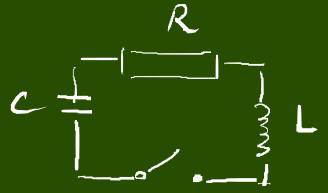
$\Rightarrow$ soluția generală: $x(t) = C_1 x_1(t) + C_2 x_2(t) = C_1 e^{\lambda t} + C_2 t e^{\lambda t}$.

$C_1, C_2 \in \mathbb{R}$.

exemplu, RLC serie $L = 1H$, $R = 2\Omega$, $C = 1F$

$Q(0) = 2C$, $I(0) = 1A$

$Q(t)$, $I(t) = ?$



ec. dif: $L \cdot Q''(t) + R \cdot Q'(t) + \frac{1}{C} Q(t) = 0$

$Q'' + 2Q' + Q = 0$

$\lambda^2 + 2\lambda + 1 = 0 \quad \lambda = -1$

$\Rightarrow$ $x_1(t) = e^{-t}$, $x_2(t) = te^{-t}$

$\Rightarrow$ $Q(t) = C_1 e^{-t} + C_2 t e^{-t} \xrightarrow{t \to \infty} 0$

$I(t) = Q'(t) = -C_1 e^{-t} + C_2 e^{-t} - C_2 t e^{-t} \xrightarrow{t \to \infty} 0$

$$\begin{cases} Q(0) = 2 \\ I(0) = 1 \end{cases} \Rightarrow \begin{cases} C_1 = 2 \\ -C_1 + C_2 = 1 \end{cases} \quad C_2 = 3$$

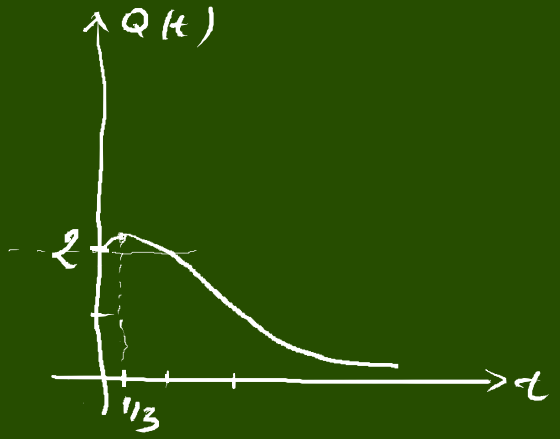
$$Q(t) = e^{-t}(2+3t)$$

$$\lim_{t\to\infty} Q(t) = 0$$

$$I(t) = Q'(t) = 0$$

$$I(t) = e^{-t} - 3te^{-t}$$

$$= \underline{e^{-t}(1-3t)} = 0 \iff t = \frac{1}{3}$$



t	0		$\frac{1}{3}$		∞
$Q'(t)$	$+$	$+$	0	$-$	$-$
$Q(t)$	↗		max	↘	

max $\quad Q\left(\frac{1}{3}\right) = 3e^{-1/3} \simeq 2{,}1$

III $\quad \lambda_1, \lambda_2 \in \mathbb{C} - \mathbb{R}$, $\lambda_2 = \overline{\lambda_1}$

Not. $\lambda = \lambda_1 = \alpha + i\beta$

$\overline{\lambda} = \lambda_2 = \alpha - i\beta$

$z(t) = e^{\lambda t}$ soluție complexă a ec. dif. $x'' + ax' + bx = 0$

$$\begin{cases} x_1(t) = \operatorname{Re} z(t) = e^{\alpha t}\cos\beta t \\ x_2(t) = \operatorname{Im} z(t) = e^{\alpha t}\sin\beta t. \end{cases}$$

$\lambda = \alpha + i\beta \Rightarrow z(t) = e^{\lambda t} = e^{\alpha t}(\cos\beta t + i\sin\beta t)$

$$\begin{cases} x_1, x_2 \text{ soluții ale ec. dif} \\ x_1, x_2 \text{ liniar indep.} \end{cases} \qquad \begin{array}{l} z'' + az' + bz = 0 \\ z = x_1 + ix_2 \quad a, b \in \mathbb{R} \\ x_1'' + ix_2'' + a(x_1' + ix_2') + b(x_1 + ix_2) = 0 \end{array}$$

$$\Rightarrow \begin{cases} x_1'' + ax_1' + bx_1 = 0 \\ x_2'' + ax_2' + bx_2 = 0. \end{cases}$$

$\Updownarrow$

$\begin{pmatrix} x_1 \\ x_1' \end{pmatrix}, \begin{pmatrix} x_2 \\ x_2' \end{pmatrix}$ liniar indep.

$\Updownarrow$

$\begin{pmatrix} x_1(0) \\ x_1'(0) \end{pmatrix}, \begin{pmatrix} x_2(0) \\ x_2'(0) \end{pmatrix}$ liniar indep. (exercițiu)

$\Rightarrow$ $x_1(t) = e^{\alpha t}\cos\beta t$, $x_2(t) = e^{\alpha t}\sin\beta t$ bază în $\mathcal{S}_0$

$\Rightarrow$ sol. generală $x(t) = C_1 e^{\alpha t}\cos\beta t + C_2 e^{\alpha t}\sin\beta t$, $C_1, C_2 \in \mathbb{R}$.

esempio. $x'' + 6x' + 13x = 0 \qquad (x(0) = 0,\ x'(0) = 0)$

$x(0) = 1, \quad x'(0) = -1$

$\lambda^2 + 6\lambda + 13 = 0 \qquad (\lambda+3)^2 + 4 = 0 \qquad \lambda_{1,2} = -3 \pm 2i$

$\alpha = -3$

$\beta = 2$

$\Rightarrow x_1(t) = e^{-3t}\cos 2t, \quad x_2(t) = e^{-3t}\sin 2t$

Sol generale $x(t) = C_1 e^{-3t}\cos 2t + C_2 e^{-3t}\sin 2t$

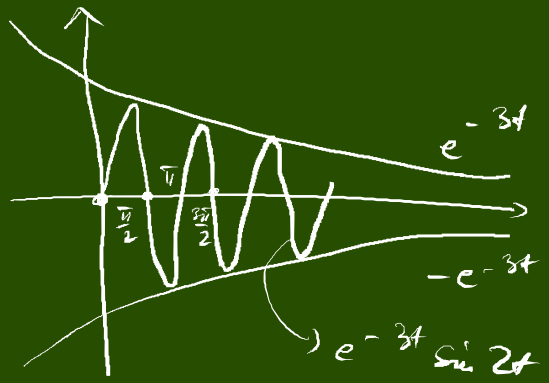
$x(0) = 1 \Rightarrow C_1 = 1$

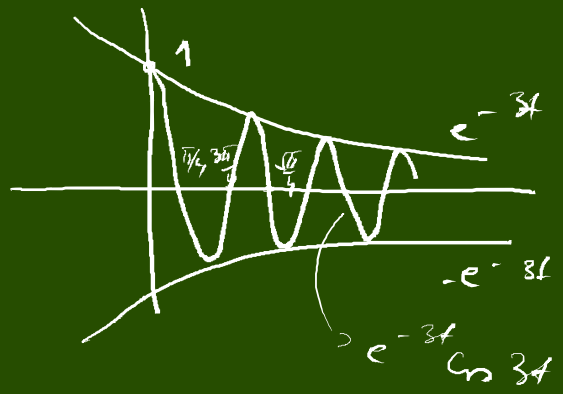
$x'(0) = -1 \Rightarrow -3C_1 + 2C_2 = -1 \Rightarrow C_2 = 1$

$x'(t) = -3C_1 e^{-3t}\cos 2t - 2C_1 e^{-3t}\sin 2t - 3C_2 e^{-3t}\sin 2t + 2C_2 e^{-3t}\cos 2t$

$\Rightarrow x(t) = \underbrace{e^{-3t}\cos 2t} + \underbrace{e^{-3t}\sin 2t} \xrightarrow{t\to\infty} 0$

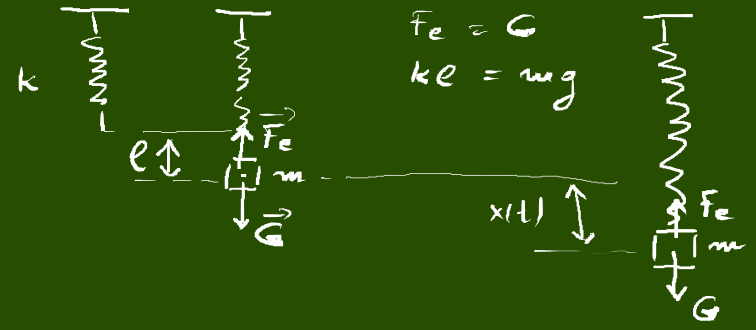
$e^{-3t}\left(\underbrace{\cos 2t + \sin 2t}_{\sqrt{2}}\right)$





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exemple :



$F_e = G$

$k\ell = mg$

$ma = mg - k(\ell + x) - F_f$

$F_f = c\,v$

$v = x'$

$a = x''$

$\Rightarrow \quad mx'' = -kx - cx'$

$x'' + \left(\frac{c}{m}\right)x' + \left(\frac{k}{m}\right)x = 0$

$2a \qquad \omega^2$

$x(0) = x_0 \quad , \quad x'(0) = v_0$

$x(t) = ?$

$$\lambda^2 + 2a\lambda + \omega^2 = 0$$

$$\Delta = 4(a^2 - \omega^2)$$

I $\Delta < 0$ $\quad 0 \le a < \omega$

I.1 $a = 0$ (fără frecare) $\quad \lambda^2 + \omega^2 = 0$, $\Delta = -4\omega^2$

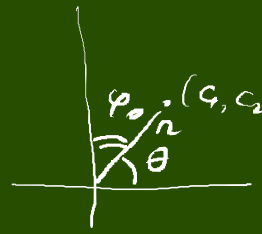
$i\omega \Rightarrow x_1(t) = \cos\omega t$

$-i\omega \Rightarrow x_2(t) = \sin\omega t$

$$\Rightarrow x(t) = \underbrace{C_1 \cos\omega t + C_2 \sin\omega t}_{\text{perioadă } \frac{2\pi}{\omega}} = A\sin(\omega t + \varphi_0)$$

frecvență $\frac{\omega}{2\pi}$

pulsație ω



A amplitudinea ?

φ_0 faza inițială ?

$$\begin{cases} C_1 = r\cos\theta \\ C_2 = r\sin\theta \end{cases}$$

$$r = \sqrt{C_1^2 + C_2^2}$$

$$x(t) = r(\cos\theta\cos\omega t + \sin\theta\sin\omega t)$$

$$= r\cos(\omega t - \theta) = r\cos(\theta - \omega t)$$

$$= r \sin\left(\underbrace{\frac{\pi}{2} - \theta}_{\varphi_0} + \omega t\right)$$

$$A = r = \sqrt{C_1^2 + C_2^2} \qquad \varphi_0 = \frac{\pi}{2} - \operatorname{arctg} \frac{C_2}{C_1}$$

$x(0) = x_0$, $x'(0) = v_0$ $\Rightarrow$ $C_1 = x_0$

$C_2 \omega = v_0$ $\Rightarrow$ $C_2 = \frac{v_0}{\omega}$.

$$\Rightarrow \quad A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$$

